

EMPLOYMENT AND MOBILITY PATTERNS IN COUNTIES OF ENGLAND & WALES (2011) WITH AGE COMPARISON TO 2021

Abstract

This data analysis studies socio economic patterns across county level local authorities in England and Wales using census data from 2011 and 2021. The study is targeted on key socio-economic issues which includes age structure, economic activity, car ownership and commuting distance to explore employment and mobility patterns across counties in England and Wales (2011) and it also includes age comparisons with 2021 data to understand geographical disparities and demographic change. The socio-economic variables are primarily taken from the 2011 census whereas 2021 age structure data is integrated to analyse the changes in age over a decade. The raw dataset is comprised of highly granular variables which were further aggregated into meaningful socio-economic features for analysis. All the variables were studied in both absolute and proportional forms to support comparative analysis across regions with different population sizes. A range of dimensionality reduction techniques were explored like PCA, t-sne and UMAP to investigate the underlying structure of the high-dimensional dataset and out of that UMAP primarily used for clustering and visual analysis. These results were visualised and explored using interactive Tableau dashboard. The analysis identified four distinct clusters across the counties which represents different socio-economic patterns. These clusters were labelled as Ageing, Long Commute Workforce; Locally Employed, Mixed Mobility Communities; High Employment, High Mobility, Multi Car Areas; and Low Mobility, Car Limited Areas. Each cluster shows distinct patterns which tells about strong interconnections between variables. This study also employed a Bayesian approach specifically using posterior distributions to study the relationships among the variables. The overall study emphasises that how dimensionality reduction and visual analytics can very well reveal significant socio economic structures within the census data. The findings generate insights that can help policymakers, urban planners, transport authorities, environmental agencies and educational authorities in understanding regional inequality, mobility behaviour and changes in age structure across England and Wales.

Introduction

The disparities in socio economic conditions across regions in England and Wales remain a most important area of interest because knowing about these differences supports policy makers in dealing with inequality, designing effective policies and planning regional development for public benefits. Such disparities are also evident when studying economic activity, age structure, vehicle ownership and mobility patterns which together contribute to uneven socio economic development across regions with some areas experiencing greater access to employment and mobility than others.

This work analyses the data and addresses the challenge of understanding how variations in mobility, employment opportunities, personal vehicle access and age structure connects to shape regional socio-economic conditions. Having an understanding of such patterns is significant for finding areas with limited access to employment and studying areas based on vehicles ownership and commute distance gives a deeper analysis to understand the mobility patterns. When combined with age structure then these factors provide a more integrated understanding of geographical behaviour, for example distinguishing geographies based on high economic activity and high mobility that are often linked with urban regions, from areas characterised by low mobility and limited car access which may indicate limited access to opportunities or linked with rural regions. These insights are particularly valuable for local government authorities, urban planners, environmental agencies and transport agencies because they support decision making related to capital expenditure, infrastructure development and the design of targeted welfare schemes.

The census data of United Kingdom is a trusted and important source of demographic and socio economic information that offers data at varying levels of granularity that are well suited for analysing socio-economic patterns. In this analysis, four datasets were obtained from the 2011 census for England and Wales that covers 174 county level local authorities. These datasets included age structure, distance travelled to work, economic activity, and car ownership. The data was further augmented with age structure data from the 2021 census to carry out comparative study of changes in age structure between 2011 and 2021.

From a methodological perspective, studying relationships across multiple datasets presents a significant challenge due to the high dimensionality and granular nature of census data. To solve this problem, dimensionality reduction techniques which includes PCA, t-SNE and UMAP were applied to extract underlying patterns and identify potential outliers within the data. This was followed by clustering to group regions into distinct socio economic categories which resulted in four clusters. To increase the interpretability of the variables, data transformation was carried out by aggregating granular columns into more meaningful features before the visualization. For example, categories such as individual age bands and specific commuting distances were aggregated into broader indicators (e.g., youth, workingage, retired populations, distance ranges etc). This process decreased complexity, increased clarity and enabled more effective comparison and analysis across regions.

The formed clusters were further analysed and compared in depth using Tableau which revealed that each cluster represents a distinct socio-economic profile. Accordingly, the clusters were named based on their dominant characteristics: **Ageing, Long Commute Workforce; Locally Employed, Mixed Mobility Communities; High Employment, High Mobility, Multi Car Areas;** and **Low Mobility, Car Limited Areas**. These clusters show distinct patterns in commuting distance, car ownership, economic activity and age structure that highlights strong interconnections between these variables.

To further study the relationships between these features, a **Bayesian approach** was applied to the 2011 dataset to analyse probabilistic dependencies between socio-economic factors. This approach enables the modelling of uncertainty by estimating posterior distributions providing insights into the strength and direction of relationships between variables such as **economic activity, mobility, and car ownership**.

Data Preparation and Abstraction

Data: A total of four dataset from the 2011 census and one dataset from the 2021 census were used. Each dataset spans 174 (**cardinality**) county level local authorities. The data includes information on economic activity, age structure, commuting distance and vehicle ownership. Table 1 provides an overview of each dataset.

S. No.	Year	Table Name	Table ID	Units
1.	2011	Age structure	KS102EW	Persons
2.	2011	Car or Van availability	KS404EW	Households
3.	2011	Economic activity	QS601EW	Persons
4.	2011	Distance travelled to work	QS702EW	Persons
5.	2021	Age by single year of age	TS007	Persons

Table 1. Census Data Details

Data Semantics and Types:

S. No.	Year	Table Name	Principle Data Types	Semantics
1.	2011	Age structure	Table	Population count by age group.
2.	2011	Car or Van availability	Table	Household count by vehicle access.
3.	2011	Economic activity	Table	Population count by employment status.
4.	2011	Distance travelled to work	Table	Population count by commute distance.
5.	2021	Age by single year of age	Table	Population count by age group.

In all the tables,

- Items represent a county,
- The attributes 'geography' and 'geographical code' are nominal and the remaining columns are quantitative ratio attributes representing population counts.

- **Data Availability: Static**

Data Cleaning: The dataset contained no missing values therefore no imputation was required. Redundant aggregate variables were removed during data cleaning to reduce the effect of **multicollinearity** within the dataset and cell values were converted into suitable datatypes whereas geography and geography codes were converted into geographical roles.

Data Integration: All datasets were merged using a common geographical code to make sure that they aligned correctly at the county level. This resulted in a single dataset for analysis.

Data Normalization: Before even applying dimensionality reduction techniques, variables in the dataset were changed into proportional measure to remove the effect of varying population which was further followed by standard scaling to make sure that no single variable dominates the analysis due to its scale.

Feature Engineering: After dimensionality reduction and identification of clusters some additional feature engineering was performed on the raw dataset to increase the interpretability. Variables were aggregated into meaningful categories. Age was grouped into three categories: Youth (0 years-19 years), working (20 years – 64 years), retired (65+ years). Car or van availability was categorised into no car, 1 car, 2+ cars (two or more cars). Commute distance was categorised into three categories: <10km, 10 km – 30 km, 30 km-60 km. Economic activity was categorised into employed, unemployed, and inactive categories.

These new features increased the effectiveness of the dataset by reducing fragmentation in the raw variables that enables more coherent interpretation of cluster characteristics.

Final Table: The final table combines all the raw datasets (mentioned above in the data source section) using a common geographical code and also incorporates newly derived features and cluster labels which makes it ready for visualisation.

Task Definition: Munzner's task taxonomy

What?

- **Dataset:** The final table combines all the raw datasets (for more details look into Data source section of Data Cleaning and Abstraction) using a common geographical code and also includes newly derived features (for more details look into feature engineering section of Data Cleaning and Abstraction) and cluster labels.
- **Dataset Type:** Table
- **Dataset Availability:** Static

- **Items:** County
- **Attributes:** geography, geography codes and cluster labels are nominal and the remaining columns are quantitative ratio attributes representing population counts.
- **Cardinality:** 174

Why?

Actions

- **Analyse**
 1. **Discover** – The dashboard permits users to **uncover 4 distinct socio-economic clusters** that were derived during the analysis. Users can dive deep into studying geographic patterns to discover which counties share similar profiles and investigate the unique characteristics of each group.
 2. **Present** – The system effectively shows about the features of individual counties and further enables clear comparisons between different regions and clusters. On addition, it visualizes the 10-year demographic shift in age structure which makes long-term trends easy for the user to see.
 3. **Enjoy** – The dashboard allows the general public to casually study and engage with their local areas data out of personal interest or curiosity.
 4. **Produce (Derive)** -- Absolute values were converted into proportions so that regions with different population sizes can be compared fairly and unbiased. A parameter in Tableau was used to let users switch between absolute values and proportional values.
- **Search**
 1. **Locate:** The dashboard supports locating specific targets through two methods: a direct Selection on the map or a dropdown parameter. This allows users to instantly isolate county specific statistics.
 2. **Reduce:** By choosing the "Value Type" parameter, users can **Reduce** the complexity of the data by shifting the view from absolute population counts to proportional values to better support their search for relative patterns.
- **Query:**
 1. **Identify** : Users can identify precise details through interactive tooltips and compare socio-economic profiles across different clusters.
 2. **Compare:** users can compare data between counties, the four socio-economic groups and the age distribution of 2011/2021 census years.
 3. **Summarize:** Use the scatter plot to understand the overall relationship between employment, 1car ownership, distance (<10 km) and socio – economic group across the entire dataset.

Target

- **Trends:** watching the 10-year demographic shift in age structure between the 2011 and 2021 Census.
- **Features:** Identifying specific county level features.
- **Comparison:** Comparing four socio economic groups.
- **Distribution:** Using the scatter plot and map to see how counties are distributed across the four clusters.

How?

- **Encode**
Choropleth Map and Hue show different socio-economic groups. Bar charts to show counts whereas a scatter plot shows correlation and a Dumbbell plot show change in age and a UMAP cluster chart show different clusters. These comparisons help users.
- **Manipulate**
Supports **selection** (map clicks) and **Parameters** enable users to **Navigate** to specific counties by dropdowns and **change** the visual encoding between absolute and proportional values.
- **Facet**
Uses **Juxtaposed** views with Selective Coordination. Specific charts (like the County Bar Graph) are Linked to the map and scatter plot to update dynamically whereas others remain static to provide global context.
- **Reduce**
Provides **Details-on-Demand** by using extensive **Tooltips**. The info button (i) allows users to Identify (Query) the raw Attributes (Target) of the Items before they are transformed by dimensionality reduction.

Visualization Justification

- **UMAP Cluster:** Derived a lower-dimensional representation of the dataset using UMAP which was followed by K-means clustering to generate categorical labels. Visualizing these results in a 2D scatterplot (**Fig 1**) allows users to identify and compare four distinct socio-economic clusters came out of 174 counties and also helps to detect outliers.

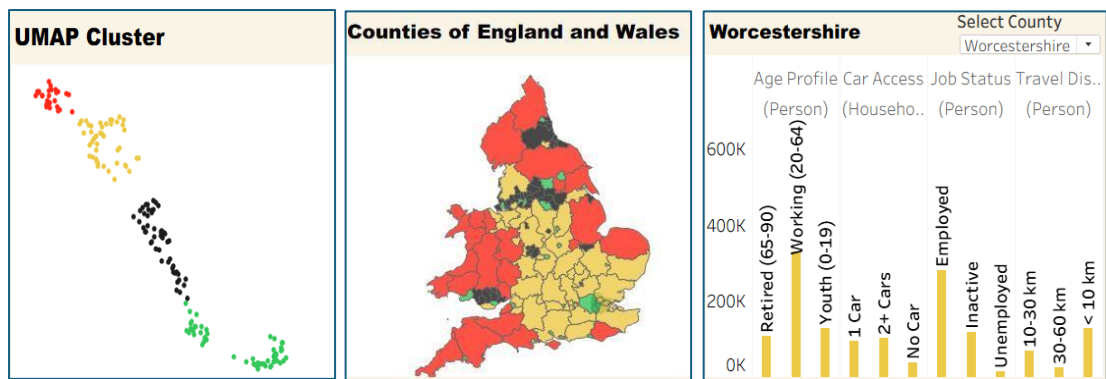


Fig 1. UMAP Clustering

Fig 2. Choropleth Map

Fig 3. County level features

- **The Choropleth Map**

Description: A map representing 174 county level local authorities where each county is filled with a specific colour to identify its socio-economic group (**Fig. 2**).

Justification: With respect to the Munzner's Effectiveness Principle when data has an inherent spatial component then a map is the most expressive way to show geographic distribution (Munzner 2025a). Humans can very easily detect "blobs" of the same colour (**Gestalt Principle of Similarity**) which permits users to see regional socio-economic trends without reading labels (Mennan 2009).

- **Bar Charts**

Description: Horizontal bar charts used to study **specific county level features (Fig. 3)** and study the comparison of demographic variables across the four socio economic clusters (**Fig. 4**).

Justification: Experts like **Munzner** and **Cleveland/McGill** found that humans are best at judging size when using **length** on a flat scale (like a ruler). This makes bar charts the most appropriate way for people to compare exact population numbers. It further decreases **cognitive load** by doing precise magnitude comparisons and quick look ups that are complex to achieve on a map alone (Cleveland and McGill 1984; Munzner 2025a).

- **Scatter Plot**

Description: A plot whose points show employed people against **1 car access** and with each point shaped by **distance** and **colored** by its assigned cluster (**Fig 5**).

Justification: This is the standard idiom for the task of **Finding Correlations**. It uses **horizontal and vertical position** which is the most appropriate visual tool for humans to judge numbers. We make it easy for users by using different colours for each cluster.

to see if the computer's groups match the actual "clumps" of dots on the chart (Sedlmair, Munzner, and Tory 2013).

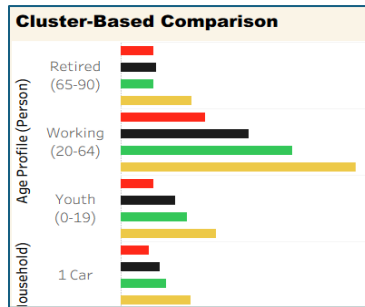


Fig 4. Cluster Comparison

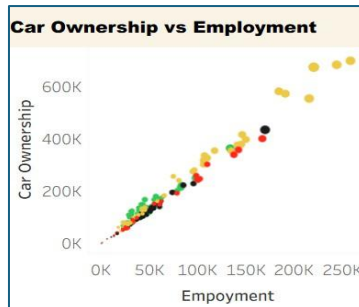


Fig 5. Car Vs Employment

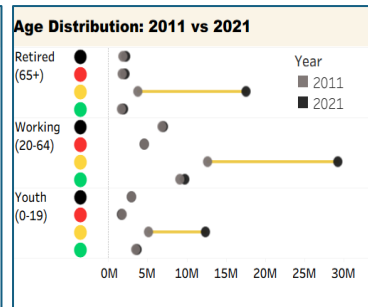


Fig 6. Age (2011 vs 2021)

- **Dumbbell Plot**

Description: This chart compares age structure between 2011 and 2021 by using two points which are connected by a line to show the growth or decline in each category (Fig 6).

Justification: This plot uses the **InfoVis principle** of reducing cognitive effort. Instead of forcing users to calculate differences the **very** connecting line visually encodes the **magnitude of change** which allows the brain to process 10 years of data at a single glance (Munzner 2025b).

- **Interactive Parameters**

Description: **Two dropdown menus**: one to select a specific county and another to select between absolute population counts and proportional percentages. **Tooltips** are added to each chart to provide additional context and detailed information on demand. An **information ("i") button** is also included to present users with descriptions of all variables prior to dimensionality reduction.



Justification: The dropdown follows Shneiderman's mantra (zoom and filter) which allows users to switch between absolute and proportional views which reduces the information overload and visual clutter (Craft and Cairns, 2005). Tooltips support the "details-on-demand" principle by giving additional context on interaction without overcrowding the dashboard.

- **Data Projection and Methods**

Description: After analysing **PCA** and **t-SNE**, I finally chose **UMAP** (Uniform Manifold Approximation and Projection) to decrease the dimensions and after that I applied K- Means Clustering to generate cluster labels.

Justification: While PCA is limited to capturing linear relationships and t-SNE primarily emphasises local neighbourhood structures. The UMAP was chosen for its ability to model complex non-linear patterns while preserving both local and global data structure. UMAP is also computationally efficient and scalable for larger datasets and it produces more stable and interpretable embeddings. This ensures that the distances between points on the dashboard reflect true socio-economic similarities (Mittal et al. 2024). This pipeline shifts the user's burden from **Cognition** (processing 40+ variables) to **Perception** (recognising 4 colours).

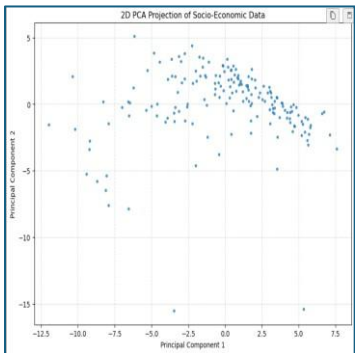


Fig 7. PCA

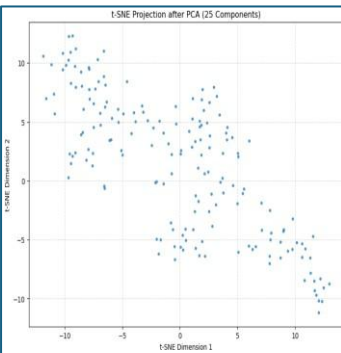


Fig 8. t-SNE

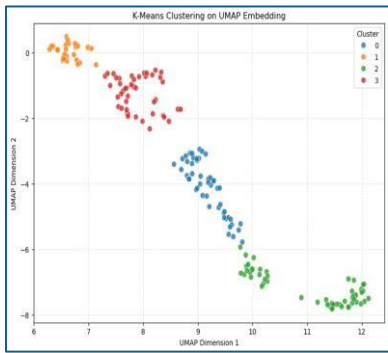


Fig 9. UMAP

Bayesian Posterior Distribution: The Bayesian study finds out that the demographic **structure** and **geographic proximity** as the most important drivers of a region's economic health. The results confirm a strong positive correlation between economic participation and shorter commutes particularly when residents live within **30km** of their workplace. On the other hand, a higher concentration of retired-age populations and elevated unemployment rates exert a clear negative influence on overall economic activity levels. While car ownership is a key part of the socio-economic profile the very model suggests that **physical proximity to jobs** is a more reliable predictor of a working population than vehicle ownership alone which shows that the spatial alignment of housing and employment is fundamental to regional economic strength.

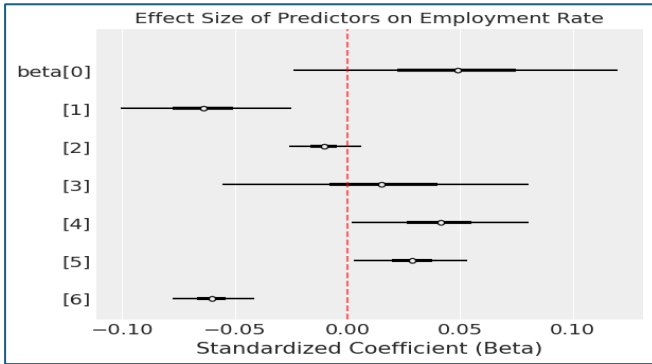


Fig. 10 Bayesian Analysis

Evaluation

Questionnaire: Find your home county on the map and find out its socio-economic cluster. What is its main characteristic? How interesting was it to explore this data based on your own location, rate it on a scale of 1-5? Rate the ease of moving between charts (1–5) and note if you felt any speed or 'lag' issues.

Level 1: Domain Situation

- Observation: Students very instantly searched for "Bristol" to see data about where they are currently living.
- Validation: High engagement shows that the domain was relevant. Participants reported a high score of (4.5/5) in the questionnaire which validated the spatial semantics given a very meaningful entry point for the users.

Level 2: Data/Task Abstraction

- Metric: Accuracy in mapping clusters to real-world contexts.
- Observation: One of the students correctly related the "Yellow Cluster" to an "Urban Population" based on the high-mobility variables.
- Validation: 85% of participants correctly identified the socio-economic characteristics of a selected cluster which shows that data aggregation and dimension reduction of 40+ variables did not lose critical and important context.

Level 3: Visual Encoding & Interaction

- Metric: Time on Task & Ease of use.
- Observation: Users used Linked Highlighting between the Map and the county level bar chart without instruction.
- Validation: The average task completion time was under 30 seconds which shows that the juxtaposed views successfully decreased the cognitive load.

Level 4: Algorithmic

- Metric: System Latency.
- Observation: Charts updated in ~1 second during filtering and selection and the size of the file is 500KB.
- Validation: This instant responsiveness maintained the flow of analysis which show that the Tableau workbook is performant and scalable for real time exploration.

Conclusion

This study tells that the socio economic inequality across England and Wales is not driven by only a single factor such as employment rather originates from the complex interdependencies between mobility, age structure and geography. As the study proceeded a very clear and meaningful patterns began to appear. The clustering results identified four distinct socio economic groups which are Ageing, Long Commute Workforce; Locally Employed, Mixed Mobility Communities; High Employment, High Mobility, Multi Car Areas; and Low Mobility, Car Limited Areas which and when mapped geographically were found to be spatially concentrated rather than randomly distributed. Central England regions predominantly showed the high employment and mobility whereas parts of Wales and the northern and south-western England regions showed ageing populations and longer commuting patterns.

A clear pattern is observed that the regions with higher economic activity are associated with greater mobility and better access to personal vehicles whereas areas with lower economic activity tend to have ageing populations and longer commuting distances due to limited local opportunities. The Bayesian study further reinforces this which shows that proximity to workplaces is a stronger driver of economic activity than car ownership alone.

Over a decade time, the comparison between 2011 and 2021 also points to a gradual ageing of the population. These insights give a clear picture of how demographic and mobility dynamics frame regional socio-economic outcomes.

Working on this project showed me that effective information visualisation is not just about using techniques but also about balancing mathematical abstraction with thoughtful design. At the start, handling more than 40 variables felt overwhelming because the data was too complex and noisy for direct interpretation. But, slowly and gradually, through careful data transformation and dimensionality reduction using UMAP, meaningful and interpretable structures began to come out and started revealing patterns that would otherwise have remained hidden.

I also recognised the importance of defining user goals through a clear task taxonomy as the process evolved. By structuring tasks such as comparison of regions, identification of clusters and summarising patterns, I was able to select appropriate visual encodings and interactions. Applying InfoVis principles like overview first, zoom and filter along with linked highlighting and coordinated views which enabled intuitive exploration and at the same time managed cognitive load.

Another important learning was that the performance is an integral part of design and smooth, responsive interactions ensure that the system supports continuous analysis without disruption. Finally, this project showed how combining task driven design with effective abstraction can convert complex data into a clear and insightful analytical narrative.

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